Wk2L2: Kirchhoff's laws & Equivalent circuits

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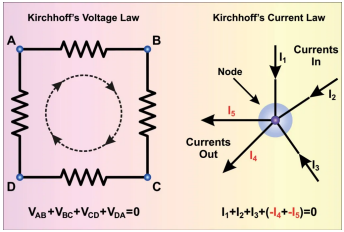

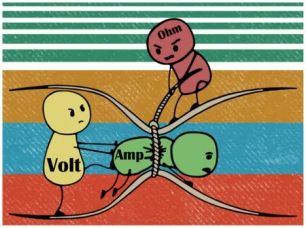

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Wed. Sept. 10

Voltage, Current, & Power

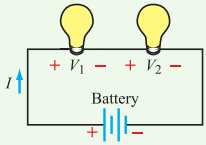
Ohm's Law

$$v = \frac{dw}{dq} = \frac{dq}{dt}R = iR \tag{1}$$

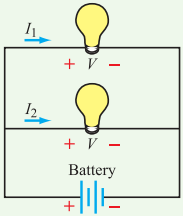
Power

$$p = \frac{dw}{dt} = \frac{dq}{dt}\frac{dw}{dq} = iv \tag{2}$$

Equivalent Resistances



(a) Series circuit



(b) Parallel circuit

Figure 1-6: Two light bulbs connected (a) in series and (b) in parallel.

Series

$$R_{eq,--} = \sum_{k=1}^{n} R_k \tag{3}$$

Parallel

$$R_{eq,||} = \left(\sum_{k=1}^{n} \frac{1}{R_k} \right)^{-1} \tag{4}$$

Voltage sources in **series**

Voltage sources in $--$

Add in series

$$v_{eq,src,--} = \sum_{k=1}^{n} v_k$$

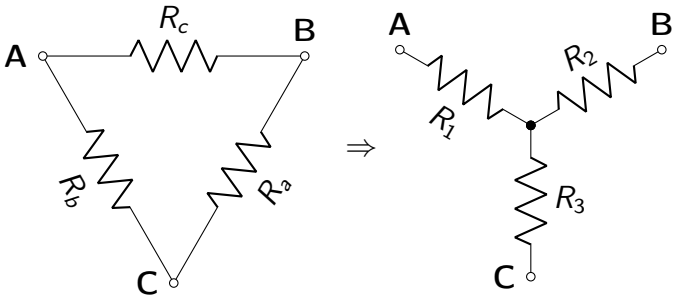
Current sources in **parallel**

Current sources in ||

Add in parallel

$$i_{eq,src,||} = \sum_{k=1}^{n} i_k$$

Delta-Wye (Δ-Y)



2-4.1 Δ →Y Transformation

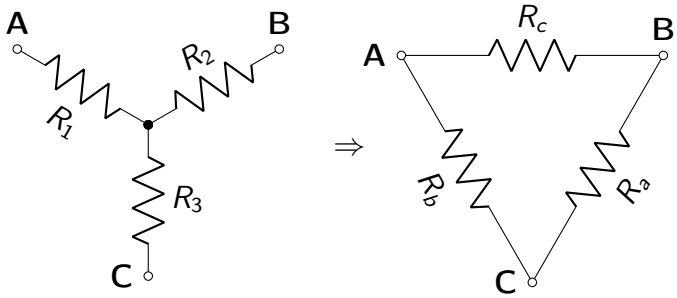
$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} \tag{2.42a}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c} \tag{2.42b}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} \tag{2.42c}$$

► R_1 of the Y circuit, which is connected to node 1, is given by an expression (Eq. (2.42a)) whose numerator is the product of the two resistors connected to node 1 in the Δ circuit, namely R_b and R_c. The same form of symmetry applies to R_2 and R_3. ◄

Wye-Delta (Y-Δ)



2-4.2 Y→ Δ Transformation

$$R_a = \frac{R_1R_2 + R_2R_3 + R_1R_3}{R_1} \tag{2.43a}$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_1R_3}{R_2} \tag{2.43b}$$

$$R_c = \frac{R_1R_2 + R_2R_3 + R_1R_3}{R_3} \tag{2.43c}$$

► R_a of the Δ circuit, which is connected between nodes 2 and 3, is given by an expression (Eq. (2.43a)) whose denominator is R_1, the resistor connected to node 1 of the Y circuit. This form of symmetry also applies to R_b and R_c. ◄

Kirchhoff's current law (KCL)

- The algebraic sum of currents entering a node must always be zero.
- The algebraic sum of currents leaving a node must always be zero.
- Total currents entering a node must equal the total currents leaving.

$$\left[\sum_{k=1}^{n} i_k\right]_{node} = 0$$

Kirchhoff's voltage law (KVL)

- The algebraic sum of voltages around a closed loop must always be zero.
- Total voltage rise around a loop must equal the total voltage drop.

$$\left[\sum_{k=1}^{n} v_k\right]_{loop} = 0$$

Table 2-4: Equally valid, multiple statements of Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL).

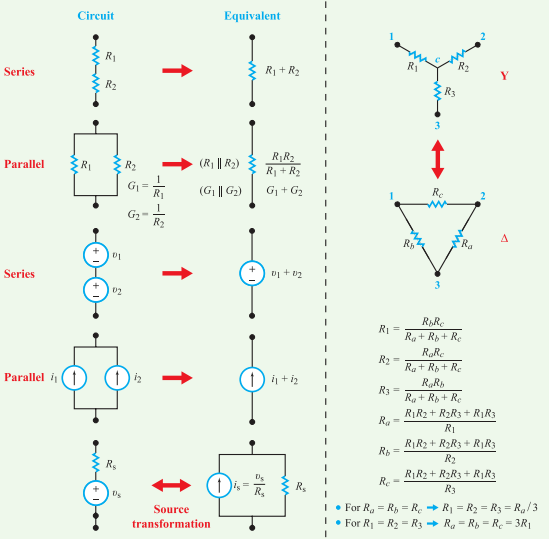
KCL

- Sum of all currents entering a node = 0
  [i = "+" if entering; i = "−" if leaving]
- Sum of all currents leaving a node = 0
  [i = "+" if leaving; i = "−" if entering]
- Total of currents entering = Total of currents leaving

KVL

- Sum of voltages around closed loop = 0
  [v = "+" if + side encountered first in clockwise direction]
- Total voltage rise = Total voltage drop

Table 2-5: Equivalent circuits.



Reminders

- HW 1 is due Monday (on paper in class)
- Recitation is Friday